Name
Date
Course

Proving Equations Using Identities (Homework)

Prove that each equation is true by converting the left side to the right.

1. $\sec A \sin A \cos A = \dfrac{1}{\csc A}$

2. $\sin^2\theta - 2 + \cos^2\theta = -1$

3. $\sec^2 y + 51 = \tan^2 y + 52$

4. $\tan W(\cot W + \tan W) = \sec^2 W$

5. $\cos^2 B(\sec^2 B - 1) = \sin^2 B$

6. $\dfrac{\sin^2\beta - \cos^2\beta}{\sin\beta\cos\beta} = \tan\beta - \cot\beta$

7. $\cos X - \cos X \sin^2 X = \cos^3 X$

8. $\dfrac{1-\tan^2\alpha}{1+\tan\alpha} = 1 - \dfrac{\sin\alpha}{\cos\alpha}$

9. $\dfrac{\cot^2\theta - 2\cot\theta - 15}{\cot\theta - 5} = \dfrac{1}{\tan\theta} + 3$

10. $\dfrac{\csc^2 A - \sec^2 A}{\csc A + \sec A} = \dfrac{1}{\sin A} - \dfrac{1}{\cos A}$

11. $\csc B - \cot B \cos B = \sin B$

12. $\tan\theta + \sec\theta = \dfrac{\sin\theta + 1}{\cos\theta}$

13. $\cot W + \dfrac{\sin W}{\cos W - 1} = \dfrac{1}{\sin W}$

14. $\dfrac{1}{1-\cot A} + \dfrac{1}{1-\tan A} = 1$

15. $\dfrac{1}{1-\sin X} + \dfrac{1}{1+\sin X} = 2\sec^2 X$

Homework Answers

★ There is more than one way to prove each equation below. Each solution demonstrates only one.

① $\sec A \sin A \cos A = \frac{1}{\csc A}$

$\frac{1}{\cancel{\cos A}} \sin A \cancel{\cos A} =$

$\sin A =$

$\frac{1}{\csc A} =$ ✓

② $\sin^2\theta - 2 + \cos^2\theta = -1$

$\sin^2\theta + \cos^2\theta - 2 =$

$1 - 2 =$

$-1 =$ ✓

③ $\sec^2 y + 51 = \tan^2 y + 52$

$\tan^2 y + 1 + 51 =$

$\tan^2 y + 52 =$ ✓

④ $\tan w(\cot w + \tan w) = \sec^2 w$

$\tan w \cot w + \tan^2 w =$

$\tan w \frac{1}{\tan w} + \tan^2 w =$

$1 + \tan^2 w =$

$\sec^2 w =$ ✓

⑤ $\cos^2 B(\sec^2 B - 1) = \sin^2 B$

$\cos^2 B \sec^2 B - \cos^2 B =$

$\cos^2 B \frac{1}{\cos^2 B} - \cos^2 B =$

$1 - \cos^2 B =$

$\sin^2 B + \cos^2 B - \cos^2 B =$

$\sin^2 B =$ ✓

⑥ $\frac{\sin^2\beta - \cos^2\beta}{\sin\beta\cos\beta} = \tan\beta - \cot\beta$

$\frac{\sin^2\beta}{\sin\beta\cos\beta} - \frac{\cos^2\beta}{\sin\beta\cos\beta} =$

$\frac{\sin\beta}{\cos\beta} - \frac{\cos\beta}{\sin\beta} =$

$\tan\beta - \cot\beta =$ ✓

⑦ $\cos x - \cos x \sin^2 x = \cos^3 x$

$\cos x(1 - \sin^2 x) =$

$\cos x(\sin^2 x + \cos^2 x - \sin^2 x) =$

$\cos x \cdot \cos^2 x =$

$\cos^3 x =$ ✓

⑧ $\dfrac{1 - \tan^2\alpha}{1 + \tan\alpha} = 1 - \dfrac{\sin\alpha}{\cos\alpha}$

$\dfrac{(1 - \tan\alpha)(1 + \tan\alpha)}{1 + \tan\alpha} =$

$1 - \tan\alpha =$

$1 - \dfrac{\sin\alpha}{\cos\alpha} =$ ✓

⑨ $\dfrac{\cot^2\theta - 2\cot\theta - 15}{\cot\theta - 5} = \dfrac{1}{\tan\theta} + 3$

$\dfrac{(\cot\theta - 5)(\cot\theta + 3)}{\cot\theta - 5} =$

$\cot\theta + 3 =$

$\dfrac{1}{\tan\theta} + 3 =$ ✓

⑩ $\dfrac{\csc^2 A - \sec^2 A}{\csc A + \sec A} = \dfrac{1}{\sin A} - \dfrac{1}{\cos A}$

$\dfrac{(\csc A - \sec A)(\csc A + \sec A)}{\csc A + \sec A} =$

$\csc A - \sec A =$

$\dfrac{1}{\sin A} - \dfrac{1}{\cos A} =$ ✓

(11) $\csc B - \cot B \cos B = \sin B$

$$\frac{1}{\sin B} - \frac{\cos B}{\sin B}\cos B =$$

$$\frac{1}{\sin B} - \frac{\cos^2 B}{\sin B} =$$

$$\frac{1-\cos^2 B}{\sin B} =$$

$$\frac{\sin^2 B + \cos^2 B - \cos^2 B}{\sin B} =$$

$$\frac{\sin^2 B}{\sin B} =$$

$$\sin B =$$

(12) $\tan\theta + \sec\theta = \dfrac{\sin\theta + 1}{\cos\theta}$

$$\frac{\sin\theta}{\cos\theta} + \frac{1}{\cos\theta} =$$

$$\frac{\sin\theta + 1}{\cos\theta} =$$

(13) $\cot W + \dfrac{\sin W}{\cos W + 1} = \dfrac{1}{\sin W}$

$$\frac{\cos W}{\sin W} + \frac{\sin W}{\cos W + 1} =$$

$$\frac{\cos W(\cos W+1)}{\sin W(\cos W+1)} + \frac{(\sin W)\sin W}{(\cos W+1)\sin W} =$$

$$\frac{\cos^2 W + \cos W}{\sin W(\cos W+1)} + \frac{\sin^2 W}{(\cos W+1)\sin W} =$$

$$\frac{\sin^2 W + \cos^2 W + \cos W}{\sin W(\cos W + 1)} =$$

$$\frac{1+\cos W}{\sin W(\cos W+1)} =$$

$$\frac{1}{\sin W} =$$

(14) $\frac{1}{1-\cot A}+\frac{1}{1-\tan A}=1$

$$\frac{(1-\tan A)}{(1-\tan A)}\frac{1}{(1-\cot A)}+\frac{1}{(1-\tan A)}\frac{(1-\cot A)}{(1-\cot A)}=$$

$$\frac{1-\tan A}{(1-\tan A)(1-\cot A)}+\frac{1-\cot A}{(1-\tan A)(1-\cot A)}=$$

$$\frac{2-\tan A-\cot A}{1-\cot A-\tan A+\tan A\cot A}=$$

$$\frac{2-\tan A-\cot A}{1-\cot A-\tan A+\tan A\frac{1}{\tan A}}=$$

$$\frac{2-\tan A-\cot A}{1-\cot A-\tan A+1}=$$

$$\frac{2-\tan A-\cot A}{2-\cot A-\tan A}=$$

$$1=$$

(15) $\frac{1}{1-\sin x}+\frac{1}{1+\sin x}=2\sec^2 x$

$$\frac{1}{(1-\sin x)}\frac{(1+\sin x)}{(1+\sin x)}+\frac{1}{(1+\sin x)}\frac{(1-\sin x)}{(1-\sin x)}=$$

$$\frac{1+\sin x}{(1-\sin x)(1+\sin x)}+\frac{1-\sin x}{(1+\sin x)(1-\sin x)}=$$

$$\frac{2}{1+\sin x-\sin x-\sin^2 x}=$$

$$\frac{2}{1-\sin^2 x}=$$

$$\frac{2}{\sin^2 x+\cos^2 x-\sin^2 x}=$$

$$\frac{2}{\cos^2 x}=$$

$$2\left(\frac{1}{\cos^2 x}\right)=$$

$$2\sec^2 x=$$